Printed Pages-3 University Roll No.....

B. Tech. (CSE) II Year IV Semester,
End Term Examination, Even Semester 2014-15
CSE224: Computer Networks

Time: 02 ½ Hours **Total Marks: 40**

Note:

1. Attempt **all** questions from Section-A, **any four** from Section-B and **any three** from Section-C.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.

Section A

Note: Attempt All Questions **(1x16=16)**

I. What are the features provided by layering in OSI Protocol Stack?

II. Mention the different physical media?

III. How does NRZ-L differ from NRZ-I?

IV. What are the steps followed in checksum generator?

V. Represent 00101100101 in Differential Manchester Encoding.

VI. What is the minimum frame time for an Ethernet with round propagation delay of 5ms?

VII. What will be the sliding window size of the sender implementing selective repeat with ARQ having the sequence numbers being represented with 4 bits?

VIII. If 64 number of subnets are to be prepared in a class B network, then what will be the length of the subnet mask?

IX. What do you mean by 10Base2?

X. What are the number of networks possible under Class C? Also mention the maximum number of hosts under each network.

XI. What is CIDR?

XII. What is Multicasting?

XIII. What are the port numbers used for FTP, Telnet, SMTP, and HTTP?

XIV. What is the difference between open-loop congestion control and closed-loop congestion control?

XV. What is the response code for a document not found client error in HTTP?

XVI. Prepare a random transposition cipher text from a plain text "GLA University".

Section B

Note: Attempt any Four Questions **(3x4=12)**

I. Explain the basic name resolution techniques in DNS.

II. Differentiate between UDP and TCP.

III. Explain the connection establishment and termination process in TCP.

IV. In a leaky bucket used to control network flow, how many bits of data are left in the bucket if the output rate is 5 kb/min, there

is an input burst of 100 kb/min for 12 s, and there is no input for 48 s?

V. Differentiate between POP and IMAP.

Section C

Note: Attempt any Three Questions **(4x3=12)**

I. At time *t*, a TCP connection has a congestion window of 4000 bytes. The maximum segment size used by the connection is 1000 bytes. What is the congestion window after it sends out 4 packets and receives acks for all of them? Suppose there is one ack per packet.

   a. If the connection is in slow-start?

   b. If the connection is in congestion avoidance (linear mode)?

II. A client uses UDP to send data to a server. The data are 16 bytes. Calculate the efficiency of this transmission at the UDP level (ratio of useful bytes to total bytes). Further, calculate the efficiency of transmission at the IP level. Assume no options for the IP header.

III. Explain the RSA Asymmetric key encryption and decryption technique with an example.

IV. Write a short note on different open-channel congestion control techniques.

Printed Pages: 03 **University Roll No.....**

End Term Examination, Even Semester, 2014-15
B.Tech. II Year, Semester-IV
CSE 222: Introduction to Microprocessor

Time: 02 ½ Hrs. **Total Marks: 40**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x16=16 Marks)**

I. How many Total T-states required for LDA instruction? Justify your answer.

II. Which instruction is used for load 16 bit data (in Hexadecimal) to register pair? Write an example?

III. If value of CS register is 2000 and DS register is 4000 and value of IP is 5050. What will be physical address generated in 8086 microprocessor for fetching an instruction?

IV. Write the values of IO/M, S1 and S0 status pins for read cycle in 8085.

V. Define subroutine and stack. How they are related to each other?

What is the use of CS pin in any chip?

VI. Which signal microprocessor is used to completely yield control of the bus to the DMA controller?

VII. What is the use of pre-fetch queue in 8086 microprocessor?

Due to which causes the microprocessors immediately terminate its present activity?

VIII. What is the addressing mode used in instruction MOV M, C?

IX. How much time required for 5 T-state, if the given operating frequency is 5 KHz? Also show the calculation?

X. How many 8259 chip can be connected in cascaded mode?

XI. Write the control word of PPI chip for IO Mode where A port work as input port in Mode 2, B port work as output port in simple Input output mode and port C as input port.

XII. Write CWR contents for PIT chip for select the counter two, Read/Load MSB only in Square Wave Generator Mode for BCD counting.

XIII. Name the two input signals of 8254 programmable timer counter.

XIV. IRR stands for...

Section B

Note: Attempt any Four Questions **(3x4=12 Marks)**

I. Elucidate the operating modes of IC which is used for input output operation with 8085 microprocessor with example.

II. What is the necessity of the programmable interval timer? Discuss the organization of IC8254 PIT with a functional block diagram.

III. Explain the parallel I/O mode of 8255?

IV. Write the assembly language program to blink port C bit 3 with 1 second delay of IC8255.

V. Write short notes on following

a) Programmed I/O b) Memory Mapped I/O

Section C

Note: Attempt any Three Questions **(4x3=12Marks)**

I. Explain idle and active cycle DMA controller with diagram. What do you understand with modes of transfer? Explain DMA transfer with block diagram.

II. Draw the internal architecture of 8255 and mention the control word format of BSR Mode.

III. Explain 8259 (programmable interrupt controller) in detail. Explain different priority modes on 8259?

IV. Describe the organization of IC8237 with block diagram.

Printed Pages: 03 University Roll No. .....

End Term Examination, 2014-15

B. Tech. II-Year, IV-Semester

CSE 223 : Computer Graphics

Time: 2½ Hours **Max Marks: 40**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section – A

Note: Attempt All Question **(1x16=16)**

I. What is the difference between computer graphics and image processing?

II. Discuss the role of frame buffer in a graphics system.

III. A Standard TV has 480 scan lines. If the aspect ratio is ¾, what is the capacity of frame buffer needed if 8 bits per pixel are used?

IV. Write main differences between Vector scan display and Raster scan display.

V. Rasterized the points of a line from (5, 5) to (13, 9) by using Bresenham's line drawing algorithm.

VI. Write pseudo code procedure to implement boundary fill algorithm.

VII. Differentiate between convex and concave polygons by using proper examples.

VIII. Write 3D matrix for rotation about X-axis.

IX. Write 3D matrix for reflection about YZ plane.

X. How many vanishing points are there in isometric projection? Explain.

XI. Explain Blobby objects with example.

XII. Write alignment matrix for a 3D object.

XIII. Write any five audio file formats.

XIV. Write the frequency range of infra sound and hyper sound.

XV. Write the name of basic illumination model.

XVI. Enlist the significance of virtual reality.

Section – B

Note: Attempt any Four Questions **(3x4=12)**

I. What is multimedia? Discuss the architecture of multimedia and also explain the applications of multimedia in detail.

II. Distinguish between image space methods and object space method for visible surface determination. Write Z-buffer algorithm for hidden surface removal.

III. What is the basic concept of hypertext? How it is different from hypermedia? Explain with proper example.

IV. Write the significance of compression. In what situation lossless and lossy compression techniques are used? Discuss in detail.

V. Generate the Shannon fano code for each symbol and compute the average number of bits required to communicate a symbol from the given table.

Symbols	Probability
A	0.15
B	0.25
C	0.30
D	0.15
E	0.05
F	0.10

Section C

Note: Attempt any Three Questions **(4x3=12)**

I. What is JPEG? Discuss all the steps for JPEG image compression.

II. Why graphics standards are required? Explain GKS with its primitives and also explain the role of metafiles in GKS.

III. What do you mean by MIDI? Discuss its file formats and how it is differing with digital audio? Explain in detail.

IV. Explain the Huffman coding technique for compression. Find the Huffman code for the following stream of data.

a,a,a,a,a,b,b,b,c,c,c,c,c,c,c,d,d,d,d,d,d,d,d,d,d,e,e,e,e,f,f

Printed Pages-2 **University Roll No.....**

B.Tech II Year IV Semester, End Term Examination, 2014-15

ITE-221: Data Base Management System

Time: 02 ½ Hours **Total Marks: 40**

Note:

1. Attempt **all** questions from Section-A, **any four** from Section-B and **any three** from Section-C.
2. Symbols have their usual meanings.

Section-A

Note: Attempt All Questions. **[1X16=16 marks]**

I. Write difference between database schema and database state.
II. What are two types of data independence?
III. What do you interpret from the following ER diagram?



IV. What is the difference between Equi join and Natural join?
V. Which are the two ways in which entities can participate in a relationship?
VI. What is entity integrity rule?
VII. In which normal form, any multi valued attributes have been removed?
VIII. What type of join is needed when you wish to include rows that do not have matching values?
IX. Write the two commands related to Data Control Language.
X. How is the HAVING clause different from the WHERE clause.
XI. Write the Armstrong's axioms.
XII. Which of the properties of dependency preservation and loss-less join must be definitely satisfied? Why?

XIII. Why intention mode locks are useful in tree based locking schemes?

XIV. What is the role of checkpoints in the database recovery?

XV. Write the protocol which restricts to employ only exclusive locks?

XVI. When is it useful to have replication or fragmentation of data?

Section-B

Note: Attempt Any Four Questions. **[3x4 = 12 Marks]**

I. What is transaction? Discuss the transaction execution state with state transition diagram.

II. What are the differences between optimistic and pessimistic concurrency control schemes? Write the three phases involved in validation based protocol.

III. Explain the properties of a transaction. Explain the usefulness of each.

IV. What do you meant by deadlock handling? Explain in brief.

V. Discuss the advantage of distributed database. Explain the data fragmentation with examples.

Section-C

Note: Attempt Any Three Questions **[4X3=12 marks]**

I. Discuss two phase locking protocol. What is the limitation of 2PL? How strict two phase locking protocol helpful in overcoming this limitation?

II. What is Log? How is it maintained? Discuss the salient features of Deferred database modification and Immediate database modification strategies in brief.

III. What do you understand by serializability of schedules? Discuss the conflict and view serializability with suitable example.

IV. Discuss two phase commit protocol for recovery management in a distributed database system.

Printed Pages: 03 University Roll No. .....

End Term Examination, 2014-15
B. Tech. II-Year, IV-Semester
CSE 221 : Operating Systems

Time: 2½ Hours **Max Marks:** 40

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section – A

Note: Attempt All Questions **(1x16=16)**

I. Why do you think that idleness in CPU occurs?

II. What are the main advantages of microkernel approach to system design?

III. What are the essential and desirable properties of an OS?

IV. Define PCB?

V. What are the different scheduling criteria?

VI. Differentiate between pre-emptive and non pre-emptive scheduling.

VII. Describe IPC?

VIII. Explain the term bounded waiting.

IX. Consider two concurrently running processes: P1 with a statement S1 and P2 with a statement S2. Suppose we require that S2 be executed only after S1 has completed. How can you implement this scheme readily by letting P1 and P2 shares a common semaphore synch, initialized to 0.

X. Explain spinlock semaphore.

XI. Explain Safe State.

XII. How a deadlock can be recovered.

XIII. Define Belady's Anomaly.

XIV. What is TLB?

XV. Define Latency and seek time with respect to disk I/O.

XVI. Define Overlays.

Section – B

Note: Attempt any Four Questions **(3x4=12)**

I. Consider the following segment table

Segment	Base	Length
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1952	96

What are the physical addresses of the following logical addresses?

i. 0, 430

ii. 1, 10

iii. 2, 500

iv. 3, 400

v. 4, 112

II. What is thrashing? What is the reason for it? How does Working-Set Model solve the problem of thrashing?

III. Describe the hardware that is required to support paging with an acceptable amount of overhead. Why are segmentation and paging sometimes combined into one scheme?

IV. Explain various types of directory structures with their merits and demerits.

V. Given a memory partitions of 100K, 500K, 200K, 300K, 600K (in order), how would each of the First-fit, Best-fit and Worst-fit algorithm place processes of 212K, 417K, 112K AND 426K (in order)? Which algorithm makes the efficient use of memory?

Section – C

Note: Attempt any Three Questions **(4x3=12)**

I. How many page faults will occur for the following reference string with 3 page frames using LRU and Optimal Page Replacement Algorithms?

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1

II. Explain the concept of multiprogramming with fixed partition along with the two versions and multiprogramming with variable partition. Explain the two methods by which the memory waste in multiprogramming with variable partition can be handled.

III. What are the basic operations of a file system? Discuss linked allocation and index allocation scheme for a file allocation and compare index allocation scheme with contiguous allocation scheme.

IV. Suppose that a disk drive has 5000 cylinders numbered 0 to 4999. The drive is currently serving a request at cylinder 143 and the previous request was at cylinder 125, the queue of pending requests in FIFO order is

86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130

Starting from current head position what is the total distance that the disk arm moves to satisfy all the pending requests for SCAN and C-LOOK disk scheduling algorithms.

Printed Pages-2 University Roll No.....

B. Tech. II Year, IV Semester, End Term Examination, 2014-15

CEE-201: Environmental Studies

Time: 02 ½ Hours Total Marks: 40

Notes:-

1. Answer all 16 questions from section A, 04 questions from section B & 03 questions from section C.
2. All questions of the particular section should be answered collectively at one place.
3. Answer should be to-the point and wherever required, be supplemented with neat sketches.

Section-A

Note: Attempt All Questions (1 x 16=16 Marks)

I. What is the importance of energy pyramid in an ecosystem?
II. Saprophytes are essential in food chain. Why ?
III. Give some examples of forest resources.
IV. Why hydrosphere is called as a heat buffer of the earth ?
V. What is rain water harvesting ?
VI. What is meant by mine drainage ?
VII. Why ozone is considered as a pollutant in troposphere ?
VIII. Differentiate weather and climate.
IX. Give any two ways for harnessing solar energy.
X. How biodiversity depletion can be solved by *ex-situ* conservation ?
XI. What are the objectives of Montreal Protocol ?
XII. Differentiate sewage and effluent.
XIII. What is the importance of CITES.
XIV. What is the purpose of Life Cycle Assessment ?
XV. How Environmental Ethics contribute to solve the environmental problems?

XVI. What are the major obstacles in the path of sustainable development in India ?

Section-B

Note: Attempt any four questions **(3 x 4=12 Marks)**

I. Discuss roles of citizens in environment protection.

II. "Environmental Economics is a powerful tool for sustainable development". Comment.

III. Write a short note on ecological footprint.

IV. "Government should play a central role in environmental protection". Why ?

V. Enumerate various efforts which are being made by NGO'S towards environmental concerns.

Section-C

Note: Attempt any three questions **(04 x 3=12 Marks)**

I. What is meant by Environment Impact Assessment ? Describe it's process in detail.

II. What are salient features of Wildlife (Protection Act), 1972. What are the penalties for violating the provisions of the Act ? Explain.

III. Explain formal and non-formal Environmental Education with suitable examples.

IV. "Environmental legislation can be made effective only by making people aware of importance of environmental laws." Justify the statement.